



ABUS: A Framework for Evaluating Protein Language Models with Multimodal Capabilities

Fatemeh Ensafitakaldani^(✉), Nurit Haspel, and Daniel Haehn

University of Massachusetts Boston, Boston, MA, USA

k.ensafitakaldani001@umb.edu

Abstract. Protein language models (PLMs) are increasingly used for tasks such as structure prediction, function annotation, and protein–protein interaction modeling. However, the rapid expansion of available models including sequence-only, structure-informed, and multimodal architectures makes it challenging for researchers to determine which model best fits a specific biological task, computational budget, or deployment setting. Existing surveys often emphasize architectural or benchmark comparisons but provide limited practical guidance for real-world usability and integration. To address this gap, we introduce the **Adaptability Bioinformatics Usability Score (ABUS)**, a structured evaluation framework that characterizes PLMs across five key dimensions: adaptability, bioinformatics relevance, usability, computational efficiency, and output suitability. Each dimension is decomposed into specific rubric-based sub-features scored on a 0/1/2 scale based on objective evidence. These sub-category values are then aggregated and transformed into a normalized overall ABUS score ranging from 0 to 100, providing both granular qualitative insights and a comparative quantitative metric. ABUS is implemented as an evidence-driven scoring engine supported by human-in-the-loop validation. The framework includes a curated database of over 20 models, an interactive web interface for exploring detailed model profiles, and a prototype AI agent designed to extract evidence and recommend scores directly from research papers and repositories.

Keywords: Protein language models · Evaluation frameworks · Structural bioinformatics · Model profiling

1 Introduction

1.1 Proteins and Protein Language Models

Proteins carry out essential biological functions through the coordinated interplay of their sequence, structure, and dynamics. Accurately modeling these relationships is central to understanding protein function, predicting interactions,

and supporting therapeutic design [1,9]. Recent advances in deep learning, particularly transformer-based architectures have enabled the development of *protein language models* (PLMs), which learn rich latent representations directly from large-scale protein sequence corpora [27,35]. These representations capture biochemical constraints and long-range dependencies, providing a foundation for tasks such as structure prediction, function annotation, variant effect estimation, and protein–protein interaction (PPI) modeling [2,22,28].

1.2 PLM Modalities and Model Families

Modern protein language models differ widely in their architectural design and input modalities. Sequence-only models, such as ESM-2 [25], rely on masked language modeling over massive protein databases like BFD [37]. Structure-aware models incorporate contact maps, distance matrices, or full 3D geometry to enhance spatial reasoning and capture long-range structural constraints [17,20]. Multimodal architectures extend these approaches by integrating additional biological signals, such as evolutionary profiles, Gene Ontology annotations, literature-derived text, or small-molecule features [4,13]. As a result, PLMs vary substantially in their bioinformatics relevance, adaptability, usability, computational requirements, and the types of outputs they produce.

1.3 Challenges in Selecting a PLM

Despite rapid progress, selecting an appropriate PLM for a given biological or computational workflow remains difficult. Benchmark studies often rely on heterogeneous datasets and evaluation protocols, limiting direct comparability across models [32,47]. Practical factors, such as documentation quality, installation complexity, licensing constraints, and hardware requirements are frequently under-reported or inconsistently documented [46]. Moreover, existing surveys typically focus on architectural descriptions rather than providing a systematic approach for assessing trade-offs between biological capability and practical usability.

1.4 The ABUS Framework

To address these limitations, we introduce the **Adaptability bioinformatics Usability Score (ABUS)**, a structured evaluation framework that profiles PLMs across five key dimensions: adaptability, bioinformatics relevance, usability, computational efficiency, and output suitability. Together, these dimensions capture both the scientific capabilities of a model and the practical considerations that influence its deployability in real-world bioinformatics workflows.

1.5 Contributions

ABUS applies a standardized 0/1/2 rubric to model attributes and aggregates these scores into interpretable category-level and overall assessments, with each

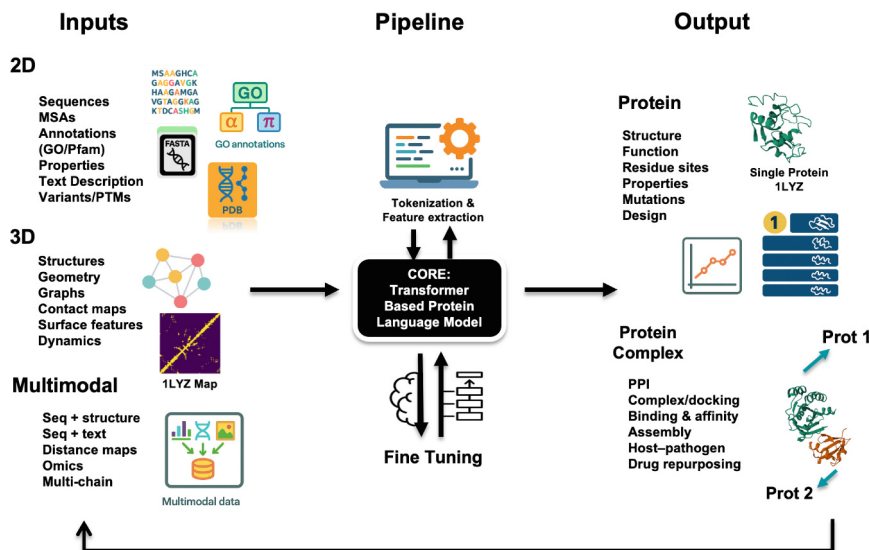


Fig. 1. Protein language models support a wide range of applications across structure prediction, function annotation, interaction analysis, and multimodal reasoning. The figure illustrates the diverse inputs these models consume such as sequences, evolutionary profiles, and structural data. The transformer-based core integrates these inputs, and creates the resulting outputs, including structures, functions, interactions, and complex biological predictions.

decision grounded in explicit evidence drawn from publications, repositories, and model cards. In this work, we make three primary contributions. First, we define the ABUS scoring formula, its five dimensions, and the associated scoring methodology. Second, we construct a curated, machine-readable database and an accompanying interactive web interface that summarize more than twenty protein language models. Third, we analyze the resulting landscape to illustrate how models vary in adaptability, biological alignment, usability, computational efficiency, and output suitability. Finally, we outline a constraint-based PLM recommender as a natural extension of the ABUS framework (Fig. 1).

2 Materials and Methods

2.1 Related Work

Protein language models (PLMs) have advanced rapidly, evolving from early sequence-only transformers to multimodal architectures that integrate structural information, evolutionary context, and literature-derived text. Fan et al. [11] survey this progression and highlight both the advantages of richer multimodal inputs and the challenges posed by sparse annotations and uneven generalization

across protein families. Recent systems such as MULAN, PoET-2, and DPLM-2 [13, 21, 40] demonstrate the effectiveness of combining structural geometry, evolutionary features, and textual descriptions to produce more bioinformatically informed PLMs.

Beyond protein modeling, the machine learning and visualization communities have developed structured evaluation methodologies that extend beyond raw performance metrics. The Scientific Visualization Future Readiness Score (FRS) [12] illustrates the value of rubric-based, evidence-driven assessment, showing that qualitative dimensions such as usability, openness, documentation quality, and composability can be evaluated systematically. Similarly, model reporting frameworks including Model Cards [29] and Datasheets for Datasets [14] emphasize transparent, evidence-grounded characterization of model capabilities and data provenance.

Benchmarking platforms such as TAPE [32], FLIP [8], and ProteinBench [47] provide standardized downstream tasks but focus primarily on performance rather than usability or practical deployment factors. In contrast, ABUS evaluates PLMs through a broader multidimensional lens, capturing model capabilities, bioinformatics relevance, usability, computational efficiency, and output suitability. To our knowledge, ABUS is the first framework to define a structured scoring scheme for PLMs, implement explicit evidence logging, and provide downstream tools including a curated database, visualization interface, and AI-assisted scoring workflow.

2.2 ABUS Framework

The Adaptability Bioinformatics Usability Score (ABUS) converts qualitative properties of PLMs into quantitative, interpretable scores. The framework has three primary components: (1) a set of categories and subfeatures capturing biological and practical dimensions of a model, (2) a standardized scoring rubric and aggregation formula, and (3) an implementation that stores both final scores and the evidence supporting them.

Categories and Subfeatures. Each PLM is evaluated across five categories designed to capture core dimensions of bioinformatics relevance and practical usability. These categories were developed collaboratively by our team of computer science and bioinformatics researchers through an iterative process of discussion and consensus, in which the final scoring categories and their corresponding weights were jointly selected to reflect their relative importance in practical research settings.

1. Adaptability reflects a model’s flexibility for fine-tuning, transfer learning, and reuse, including checkpoint availability, licensing, and support for parameter-efficient techniques such as LoRA [16] and prefix tuning [26].

2. Bioinformatics Relevance measures alignment with key biological data modalities (sequence, structure, evolutionary information) and suitability for tasks such as structure prediction, PPI modeling, and function annotation [13, 22, 28].

3. Usability captures practical barriers to adoption, including documentation quality, code maturity, installation complexity, and community engagement. This aligns with reproducibility literature emphasizing transparency and ease of use [30].

4. Computational Efficiency assesses parameter count, memory footprint, inference cost, and deployability on commonly available research hardware.

5. Output Suitability evaluates interpretability, residue-level resolution, compatibility with downstream workflows, and adherence to explainability principles in machine learning [10].

Each category consists of multiple subfeatures scored on a three-point rubric: 0 (unsupported), 1 (partially supported or ambiguous), and 2 (fully supported with clear evidence). ABUS also includes a dedicated subfeature for benchmark clarity under the *Usability* category, which captures whether a model’s evaluation setup, datasets, and reported metrics are clearly described and reproducible. However, unlike benchmarking frameworks such as TAPE [32] or FLIP [8], ABUS does not incorporate or aggregate heterogeneous downstream performance values into its scores.

Scoring Formula. Let $s_f \in \{0, 1, 2\}$ denote the score assigned to subfeature f within category c , and let w_f be its weight. For each category c with subfeature set F_c , the category score is computed as a weighted average:

$$S_c = \frac{\sum_{f \in F_c} w_f s_f}{\sum_{f \in F_c} w_f}. \quad (1)$$

Let C denote the set of ABUS categories and w_c the weight assigned to category c . The overall and normalized ABUS scores are then:

$$\text{ABUS Score} = \sum_{c \in C} w_c S_c, \quad \text{ABUS}_{\text{norm}} = \frac{\text{ABUS Score}}{\sum_{c \in C} w_c} \times 100. \quad (2)$$

By default, ABUS weights emphasize bioinformatics relevance (30%), followed by adaptability and output suitability (20% each), and usability and computational efficiency (15% each), consistent with usage patterns reported in PLM surveys [46]. These weights are configurable to support task- or application-specific evaluation profiles (Fig. 2).

Evidence-Grounded Manual Scoring. ABUS scores are assigned through a manual, evidence-driven review performed by our research team. For each model, reviewers extract supporting evidence from papers, repositories, configuration files, and model cards [29]. AI-assisted tools (e.g., ChatGPT and Gemini) are used to surface relevant passages, cross-reference findings with other AI agent systems, check consistency, and verify claims, but all scoring decisions are ultimately made by human evaluators. Following a conservative rubric [12], fully documented capabilities receive a score of 2, partially supported or ambiguous

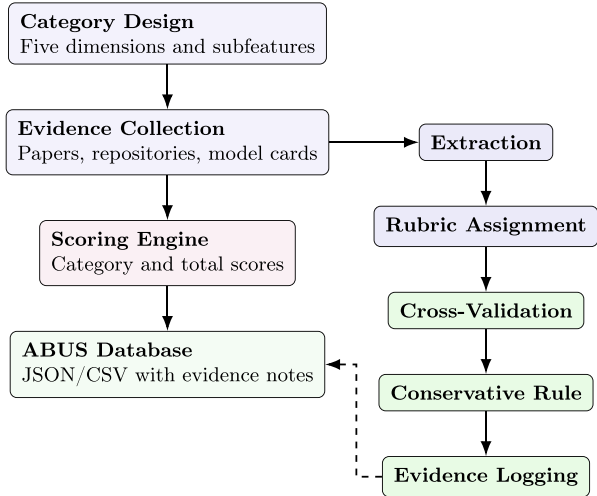


Fig. 2. Overview of the ABUS evaluation workflow. The process begins with defining category dimensions, followed by systematic evidence collection and extraction. Human reviewers assign rubric-based scores, cross-validate findings using conservative rules, and log all evidence into a structured ABUS database.

claims receive 1, and undocumented features receive 0. All evidence notes are stored in structured, machine-readable form, and we also release the protocols and prompts used for AI-assisted evidence collection on GitHub to ensure transparency and reproducibility.

2.3 Data Collection and Implementation

We curated over twenty representative PLMs, including general-purpose sequence models, multimodal systems, structure-aware architectures, and specialized predictors. For each model, we extracted metadata covering modalities, supported tasks, licensing, checkpoint availability, dependencies, implementation, and benchmark documentation. Information sources included peer-reviewed publications, preprints, GitHub repositories, model cards, and aggregation platforms such as Hugging Face [42] and the Language Model Zoo [5] [6]. This curation process follows established practices for documenting biological foundation models [49].

2.4 Preprocessing and Database Construction

Metadata and evidence were processed using a standardized pipeline inspired by dataset documentation frameworks such as Datasheets [14]. The pipeline harmonized task names, input modalities, and metric terminology, and validated inconsistencies according to ABUS’s conservative evidence rules [12]. Final structured records were exported as version-controlled JSON and CSV files forming

the ABUS database. A web interface renders radar plots, PCA projections, score distributions, and model summaries, following visualization conventions used in PLM evaluation [33].

2.5 Evaluation Setup

All models in the curated set were evaluated using the complete ABUS rubric. Reviewers inspected at least one primary publication and one implementation artifact per model, documented evidence for every sub-feature, and computed category-level and normalized ABUS scores using the formulas in Sect. 2.2. To analyze global patterns and variability across models, we computed descriptive statistics, generated radar charts, and produced PCA embeddings, similar to approaches used in prior PLM comparison studies [4, 13].

3 Results

3.1 Category-Level Patterns

Table 1 summarizes the aggregated ABUS category scores for all evaluated models. Multimodal and structure-aware systems such as MULAN [13], PoET-2 [21], DPLM-2 [40], HelixProtX [7], and EquiBind [36] achieve high bioinformatics relevance by explicitly modeling structure and interactions. General-purpose PLMs including ProCyon [31], Evola [51], ProtLLM [52], and ProteinChat [38] tend to perform well in adaptability and output suitability, but display more variability in bioinformatics relevance. Open-source pipelines and ensembles such as MetaDegron [50], FAPM [44], and DeepDrug [48] tend to show stronger usability signals in our rubric, reflecting relatively clearer implementation artifacts, while others such as Prot2Chat [41] remain more mixed on usability despite strong capability scores. Task-specialized architectures like TUnA [23] and VisionTransformer-PPI [18] attain strong output suitability by aligning directly with PPI or docking scenarios.

Notable models such as EquiBind [36], DeepDrug [48], PoET-2 [21], and AlphaFold 2 [22] combine strong bioinformatics relevance with good output suitability and at least moderate usability. Systems like MULAN [13], DPLM-2 [40], and Prot2Chat exhibit relatively balanced profiles across ABUS dimensions are generally strong across categories, with some dips in usability/efficiency depending on the model. In contrast, models such as ProCyon and Evola highlight trade-offs in usability or computational efficiency despite strong adaptability and bioinformatics relevance. HyboWaveNet achieves moderate overall performance driven primarily by task alignment, but shows relatively low biological input richness and limited efficiency, reflecting its specialization for a network-based PPI setting rather than broad multimodal bioinformatics modeling. Because model repositories, checkpoints, and capabilities are frequently updated, ABUS scores are maintained as a living resource, and values may be revised as new evidence, versions, or implementations become available.

Table 1. ABUS category and total scores for evaluated protein language models across Adaptability, bioinformatics Relevance, Usability, Computational Efficiency, and Output Suitability. Category scores are on a 0–2 scale, and the reported $ABUS_{norm}$ score is the normalized weighted sum on a 0–100 scale.

Model	Adaptability	Bio. relevance	Usability	Computational efficiency	Output suitability	$ABUS_{norm}$
MULAN	2.00	2.00	1.00	1.50	1.50	83.8
ProCyon	2.00	2.00	0.67	0.50	2.00	78.8
EvoLa	2.00	1.67	0.67	1.00	1.67	74.2
PoET-2	2.00	2.00	1.33	2.00	1.67	91.7
DPLM-2	2.00	2.00	1.33	1.00	1.50	82.5
Prot2Chat	2.00	2.00	0.67	1.50	1.50	81.3
MetaDegron	1.50	2.00	1.00	0.50	1.50	71.3
FAPM	1.50	1.00	1.33	0.00	1.50	55.0
ProtST	1.50	2.00	0.67	1.00	1.50	72.5
HelixProtX	2.00	1.50	1.00	0.50	1.50	68.8
S2F	1.50	2.00	0.33	1.50	1.00	68.8
AFTGAN	1.50	1.50	1.33	1.00	1.50	70.0
CAT-CPI	1.50	1.50	0.67	1.00	1.50	65.0
DeepDrug	2.00	2.00	1.33	1.00	1.50	82.5
TUnA	1.50	1.00	1.33	1.50	1.50	66.3
HyboWaveNet	1.50	1.00	1.33	0.50	1.50	58.8
MM-StackEns	1.50	1.40	0.50	1.00	1.33	58.8
EquiBind	1.50	2.00	1.33	1.50	1.50	81.3
VisionTransformer-PPI	1.50	2.00	0.67	1.00	1.50	72.5
RF-Ensemble	1.00	1.00	1.00	2.00	2.00	67.5
AlphaFold 2	1.50	2.00	1.33	0.50	1.50	73.8

3.2 Trade-off Analysis

ABUS profiles make cross-model trade-offs explicit. Table 2 summarizes normalized category scores for three representative multimodal PLMs, illustrating how similar biological capabilities can be paired with different usability and efficiency characteristics.

Table 2. Category-level ABUS scores (0–100) for three representative models, illustrating different trade-off profiles.

Model	Adaptability	Bio. Relevance	Usability	Comp. Efficiency	Output Suit.
PoET-2	100	100	66.5	100	83.5
MULAN	100	100	50.0	75.0	75.0
DPLM-2	100	100	66.5	50.0	75.0

MULAN [13] emphasizes multimodal bioinformatics relevance but offers only moderate usability and computational efficiency. PoET-2 [21] combines strong bioinformatics relevance with high computational efficiency and relatively high usability, making it attractive in resource-limited settings. DPLM-2 [40] provides

a balanced profile with high adaptability and bioinformatics relevance but lower computational efficiency, which may be acceptable when hardware constraints are less stringent. ABUS makes these trade-offs explicit and supports model choice based on task and resource requirements.

3.3 PCA Analysis of ABUS Profiles

To explore relationships between models within the five-dimensional ABUS space, we applied Principal Component Analysis (PCA) to the vector of category scores (adaptability, bioinformatics relevance, usability, computational efficiency, and output suitability). PCA is widely used in PLM evaluation studies [4,33] and provides a compact two-dimensional embedding that highlights the global structure across systems (Fig. 3).

The first two principal components explain the majority of the variance in the five-dimensional ABUS space. PC1 reflects overall capability strength, with the highest positive loadings arising from bioinformatics relevance, output suitability, and adaptability. In contrast, PC2 serves as an axis that separates models

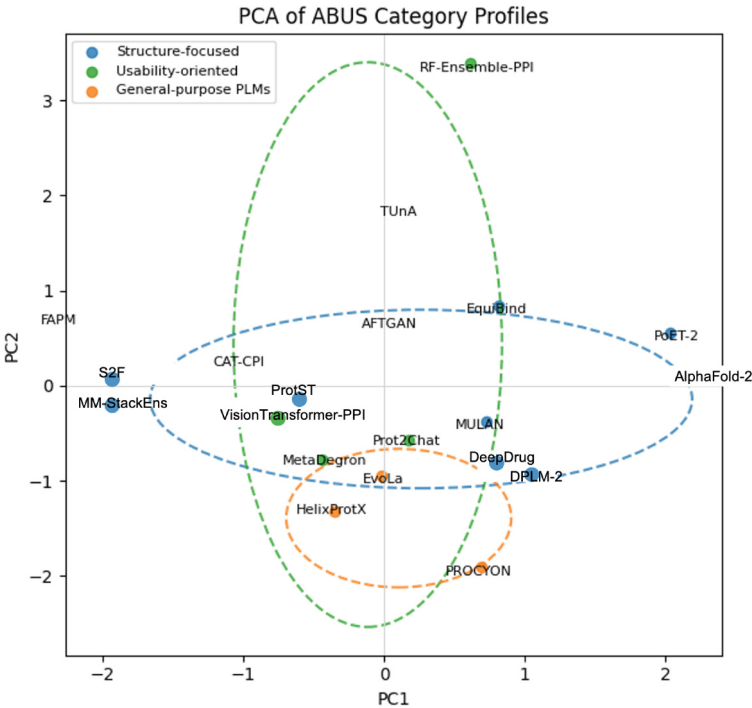


Fig. 3. Models are positioned based on their ABUS scores, revealing three natural groups: structure-focused models, pipelines with relatively strong usability or accessibility characteristics, and general-purpose PLMs. Ellipses highlight how models cluster according to their capability profiles.

emphasizing usability and computational efficiency from those more strongly focused on structural and interaction modeling, due to opposing feature contributions.

Three qualitative groups emerge in the PCA projection. First, a cluster of structure-focused and interaction-focused models appear, comprising MULAN [13], DPLM-2 [40], HelixProtX [7], EquiBind [36], and AlphaFold 2 [22]. A second group contains pipelines with relatively stronger usability or accessibility characteristics, including MetaDegron [50], ProtST [45], DeepDrug [48], and Prot2Chat [41]. Finally, a general-purpose PLM cluster is formed by Pro-Cyon [31], Evola [51], ProtLLM [52], and ProteinChat [38].

Taken together, the PCA reveals a continuous landscape ranging from multimodal, structure-aware PLMs to more usability-focused pipelines and general-purpose sequence models. The projection also highlights that relatively few systems combine very high usability with deep multimodal bioinformatics relevance, pointing to opportunities for future PLM development (Fig. 4).

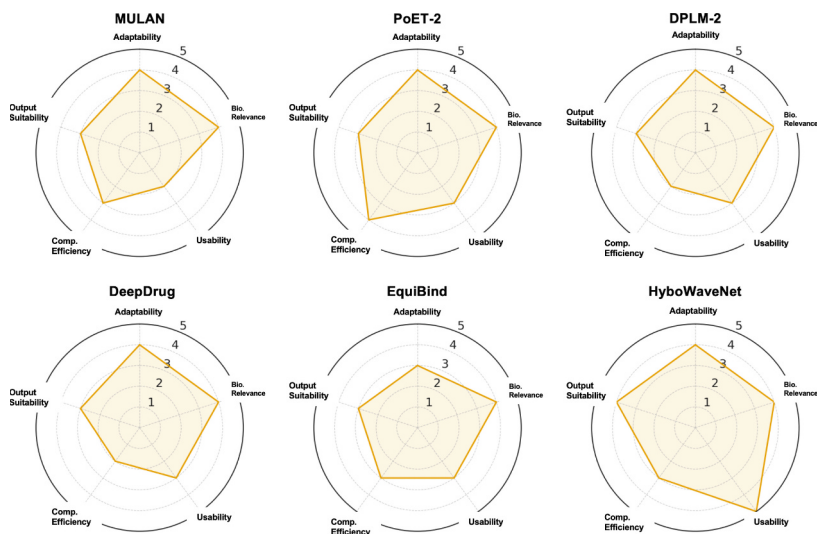


Fig. 4. Radar plots of ABUS category scores for a subset of models. Each polygon highlights the strengths and weaknesses of an individual model across the five ABUS dimensions.

4 Conclusion

We presented the **Adaptability Bioinformatics Usability Score (ABUS)**, a structured and evidence-based framework for evaluating protein language models. ABUS defines five meaningful categories, applies a standardized 0/1/2 rubric, and aggregates scores into interpretable category-level and overall assessments.

In addition, we developed a curated ABUS database, a lightweight web interface for interactive exploration, and a prototype AI agent that assists with evidence extraction and preliminary scoring, an approach supported by recent work showing that large language models can reliably aid annotation and information extraction [15, 39]. Applying ABUS to more than twenty modern PLMs shows that models occupy distinct regions of the evaluation space: some emphasize multimodal bioinformatics relevance, others prioritize usability or output suitability, and others achieve strong computational efficiency. Rather than identifying a single best model, ABUS provides a principled way to reason about trade-offs and select models aligned with specific tasks or resource constraints.

The ABUS framework, database, and visualization code are available at <https://github.com/kattens/ABUS>.

4.1 Discussion

ABUS encourages viewing protein language models as *multidimensional capability profiles* rather than as single scalar metrics, aligning with principles of structured model documentation and evaluation [14, 29]. By decomposing evaluation into adaptability, bioinformatics relevance, usability, computational efficiency, and output suitability, the framework helps practitioners determine which models fit structure-aware workflows, which are best suited for rapid prototyping or teaching, and where gaps remain in the current model landscape.

The ABUS database and accompanying visual interface make these profiles accessible, enabling comparison across systems and exploration of broader model clusters. While the current scores are verified through a human-in-the-loop process to ensure high fidelity, the framework is designed as a scalable, evidence-driven engine. By leveraging an AI agent to extract technical evidence and recommend scores directly from research repositories, ABUS minimizes subjective bias and addresses the challenge of keeping up with the rapid evolution of the field. This hybrid approach—combining automated extraction with expert validation—complements benchmark-driven evaluations by incorporating usability, efficiency, and multimodality into a unified, transparent framework, aligning with broader trends in human-centered system assessment [3].

4.2 Future Work and Research Vision

Recommender Systems. A natural next step is a constraint-aware recommender built on the ABUS database. Recommender-system paradigms [34] are well-suited for filtering models, ranking candidates using ABUS or custom weights, and visualizing trade-offs through interactive components. Developing and evaluating such a recommender, including user studies, is left for future work.

Automated scoring with LLMs. Our prototype scoring agent suggests that large language models can assist in evidence extraction and rubric assignment

[15,39]. Future extensions include formal agreement studies with expert labels, domain-specific fine-tuning, and a standalone agent enabling external groups to maintain ABUS-style databases.

Database Expansion and Community Contributions. The ABUS database will be extended as new models are released and opened to community submissions via version-controlled pull requests. In addition, researchers will be able to use the scoring system and submit their own papers through our website portal to receive preliminary ABUS evaluations. Approved submissions and their corresponding scores will be incorporated into the database, further supporting continuous expansion. Variants tailored to specific tasks (e.g., ABUS-PPI or ABUS-Structure) can be supported through configurable weights and expanded subfeatures.

Validation and Extension. We plan to conduct several follow-up studies, including a user agreement study to assess inter-rater reliability, an evaluation of practitioner feedback, and a sensitivity analysis of category weights. These experiments aim to further validate the robustness and practical relevance of the ABUS framework.

Beyond Protein Modeling. The ABUS category-based scoring framework with explicit evidence logging and customizable weights is broadly applicable to other scientific foundation models, including genomics [19], chemistry [43], and climate modeling [24]. In these domains, issues of usability, computational cost, and multimodal inputs are similarly central, positioning ABUS as a general template for structured, evidence-grounded model evaluation.

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